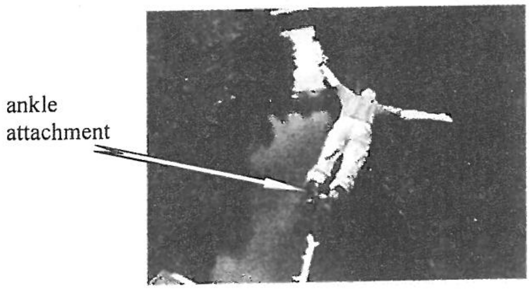
6. Read the following description about '**Bungee jumping**' and answer the questions that follow.

Bungee jumping is an activity that involves jumping from a tall structure while the person is connected to it via a thick elastic cord. When the bungee jumper jumps, the cord stretches after falling a certain distance. The bungee jumper is momentarily at rest at the lowest point but then bounces back up into the air. The bungee jumper continues to oscillate up and down a few times before he comes to a complete stop.



A simple 'ankle attachment' (as shown in the above photo) can be used to secure the player to the cord. However, due to accidents where the ankle attachment became detached from the bungee jumper, many operators now use a 'full body harness'.

When answering the following questions, neglect the effects of air resistance.

(a) (i) Describe the acceleration of the bungee jumper during the first downward fall to the lowest point. (3 marks)

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9. Carbon-14 dating can be used to identify the age of some objects which have the ^{14}C isotope, as it is radioactive and decays by emitting a β-particle. A piece of wood sample is examined using carbon-14 dating and its activity is 0.2 Bq. The half-life of ^{14}C is 5730 years. Given: 1 year $= 3.16 \times 10^7$ s

*(a) Calculate the decay constant of ^{14}C in s^{-1}. Hence find the number of ^{14}C nuclei in this wood sample. (3 marks)

Assume that living organisms contain a constant proportion of carbon-14 in the ratio of $^{14}C / ^{12}C = 1.3 \times 10^{-12}$ during its life time via intake of carbon dioxide (CO_2) from the atmosphere.

(b) The carbon content of this wood sample is found to contain a total of 1×10^{23} carbon nuclei. Estimate the number of ^{14}C nuclei in the sample originally when it died. (1 mark)

*(c) Estimate the age of this wood sample in years using the results found in (a) and (b). (2 marks)

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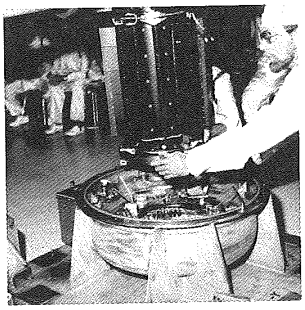
10. *Voyager I* is a space probe designed by NASA to operate for over ten years in space. It was equipped with a radioisotope thermoelectric generator (RTG) which can convert the energy released from the decay of a radioactive source into electrical power. *Voyager I* operates with a plutonium-238 radioactive source that undergoes α-decay.

(a) The plutonium-238 source is sealed inside a thin metallic casing of the RTG. The photo shows a NASA staff handling the RTG with his bare hands. Explain why it is fine for it to be handled by the staff in this way. (1 mark)

Figure 10.1



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When *Voyager I* was launched, the number of plutonium-238 atoms in the source was 3.2×10^{25}.
Given: half-life of plutonium-238 = 87.74 years.
Take 1 year = 3.16×10^{7} s.

(b) *(i) Find the activity, in Bq, of the plutonium source at the time of launch. (3 marks)

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(b) (ii) When a plutonium-238 atom decays, it releases 5.5 MeV of energy. Estimate the power, in kW, delivered by the source at the time of launch. (2 marks)

*(iii) The RTG of *Voyager I* is still in operation as *Voyager I* just left the solar system in September 2013 after it was launched 36 years ago. Estimate the corresponding power delivered by the plutonium source, expressed in **percentage** of the power delivered at the time of launch. (2 marks)

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10. (a) Part of the decay series of radium-226 (Ra-226) is shown below. Ra-226 decays to radon (Rn) by emitting an α particle with half-life 1600 years. The end product in the series is lead (Pb), which is stable.

$$^{226}_{88}\mathrm{Ra} \xrightarrow{\alpha} \mathrm{Rn} \longrightarrow \ldots \longrightarrow \mathrm{Pb}$$

(i) $^{206}_{82}\mathrm{Pb}$, $^{207}_{82}\mathrm{Pb}$ and $^{208}_{82}\mathrm{Pb}$ are three stable isotopes of lead. State, with a reason, which isotope can be the end product in this series. (2 marks)

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*(ii) In a certain laboratory, a Ra-226 source has been used for 50 years. Estimate the percentage of undecayed Ra-226 left after this period. (2 marks)

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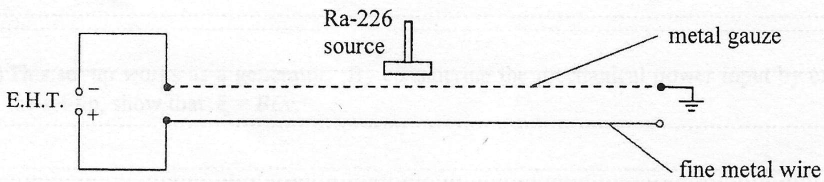
(b) Spark counters can show the ionizing power of radiations. Figure 10.1 indicates the main features of a type of spark counters in school laboratories.

Figure 10.1



A spark counter consists of a fine metal wire mounted a few mm beneath an earthed metal gauze. The wire is connected to the positive terminal of an E.H.T. (Extra High Tension) supply so that a very intense electric field is set up between the wire and the metal gauze. When a Ra-226 source is brought near the gauze, sparks giving out flashes of light and crackling sound are produced at irregular intervals.

(i) Explain why the sparks occur at **irregular** intervals. (1 mark)

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A Ra-226 source used in school laboratories is usually said to emit α, β as well as γ radiations.

(ii) Explain why β radiation is also emitted even though the source is primarily an α-emitter. (1 mark)

(iii) Why is the sparking mainly caused by α radiation rather than β or γ radiation ? Suggest a simple way to verify this. (2 marks)

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9. Potassium-40 ($^{40}_{19}K$) is a natural radioisotope of potassium.

(a) (i) What kind of decay does $^{40}_{19}K$ undergo if it decays to $^{40}_{20}Ca$? (1 mark)

(ii) As banana is rich in potassium, a student claims that the radiation emitted by $^{40}_{19}K$ after eating a few bananas can be detected outside the human body. Explain whether this claim is justified. (1 mark)

*(b) A banana typically contains 0.45 g potassium in which 0.012% by mass is $^{40}_{19}K$ while the rest is $^{39}_{19}K$.

Given: half-life of $^{40}_{19}K = 1.25 \times 10^9$ years

1 year = 3.16×10^7 seconds

molar mass of $^{40}_{19}K$ = 40.0 g

(i) Estimate the number of moles of $^{40}_{19}K$ in a banana. (1 mark)

(ii) Deduce the activity, in Bq, of a banana. (2 marks)

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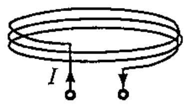
12. Figure 12.1 shows a coil of several turns.

Figure 12.1



(a) Sketch the magnetic field pattern produced by the coil when it carries a current I as shown. (2 marks)

(b) As shown in Figure 12.2(a), this coil is now connected to an a.c. source to serve as a transmitter coil (T). A receiver coil (R) connected to a CRO is placed directly above T. The time variation of the voltage of coil R is displayed on the CRO (Figure 12.2(b)).

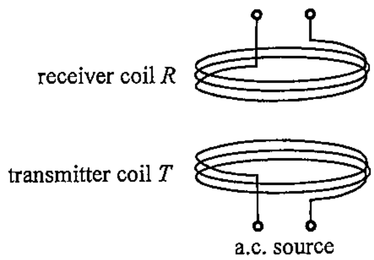


Figure 12.2(a)

CRO display

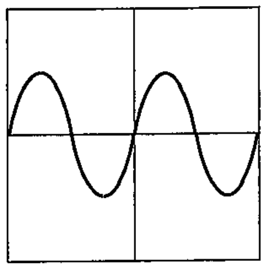


Figure 12.2(b)

(i) Explain how the voltage of the receiver coil R is produced. (2 marks)

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*(ii) On Figure 12.2(b), sketch the CRO display when the frequency of the a.c. source is doubled. (2 marks)

(c) A mobile phone with a receiver coil inside can be charged wirelessly by placing it on a charging pad equipped with a built-in transmitter coil, which is connected to a power source.

Explain why wirelessly charging of phones need phone cases with non-metallic back covers. (2 marks)

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13. Radon (Rn-222) is a radioactive gas with no colour, no smell and no taste. It is released from bedrock material underground or from concrete composed of granite. Rn-222 decays with a half-life of 3.82 days by giving out α particles.

(a) *(i) Find the decay constant (unit: s^{-1}) of Rn-222. (2 marks)

(ii) Inside buildings, the average activity due to radon in 1 m^3 of air is about 48 Bq. Estimate the average number of radon atoms in 1 m^3 of air inside buildings. (2 marks)

(iii) Why is indoor radon considered harmful to human although α particles have limited penetrating power ? (2 marks)

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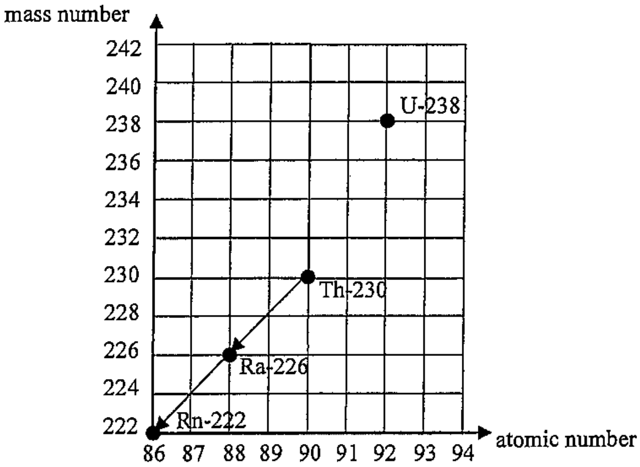
(b) Rn-222 belongs to a decay series starting from uranium-238 (U-238). It is known that U-238 undergoes decay in the sequence α-β-β-α to become thorium-230 (Th-230). Complete the decay series below from U-238 to Th-230 using arrows. (2 marks)



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12. Carbon-14 ($^{14}_{6}C$) nuclide decays into stable nitrogen-14 ($^{14}_{7}N$) nuclide with a half-life of 5730 years. Assume that the concentration of carbon-14 in the atmosphere remains unchanged over time.

(a) (i) Write an equation to represent the decay of carbon-14. (1 mark)

*(ii) Find the decay constant (in yr^{-1}) of carbon-14. (2 marks)

*(iii) 1 g of an ancient wood sample has a carbon-14 activity that gives a count rate of 6.0 counts per minute. Estimate the age of this sample in years. Given that the carbon-14 activity in 1 g of fresh wood gives a count rate of 13.6 counts per minute. (2 marks)

(b) A more advanced method involves directly measuring the number of carbon-14 atoms relative to that of the carbon-12 atoms. Carbon atoms from the wood sample are ionized to form a beam which is directed into a magnetic field as shown in Figure 12.1. Each of the positive carbon-14 and carbon-12 ions possesses one electronic charge in magnitude.

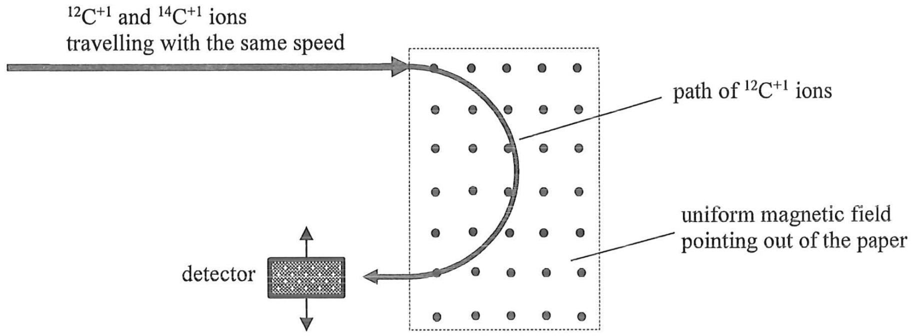

Figure 12.1

(i) Both carbon-14 and carbon-12 are isotopes of carbon. What is meant by 'isotopes'? (1 mark)

*(ii) Sketch the path of $^{14}C^{+1}$ ions on Figure 12.1. (1 mark)

(iii) Explain why the speed of the ions remains unchanged after passing through the magnetic field. (2 marks)

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